

INTRODUCTION TO DATA SCIENCE

Data Fundamentals

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- 1 Data Science Basic Concepts
- 2 What is to be a Data Scientist



Outline

- 1 Data Science Basic Concepts
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HISTORY OF DATA



1600s

John Graunt introduces the concept of data analysis in 1663.

1800s

Herman Hollerith designs a machine that helped complete the US census in 1890.

1900s

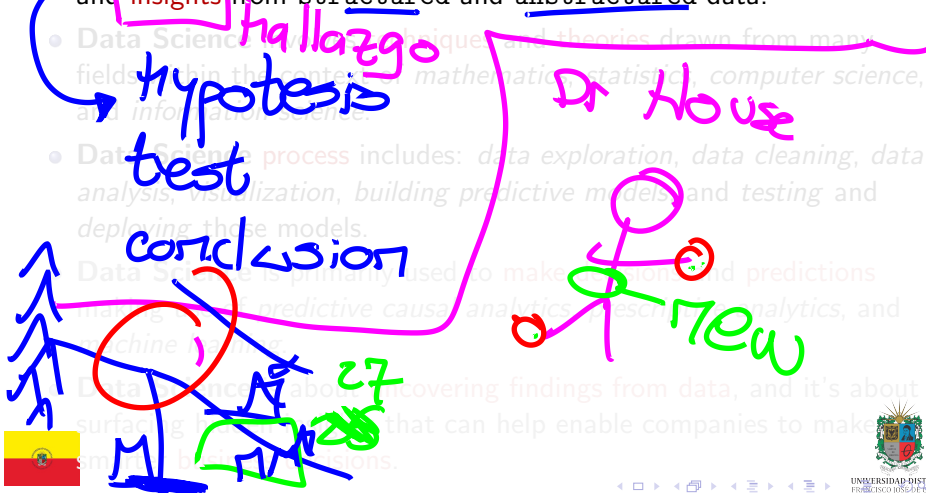
Fritz Pfleumer invents the magnetic tape which later inspired the invention of floppy disks and hard disk drives.

1990s

Sir Tim Berners Lee
invents the
World Wide Web.

What is Data Science?

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- **Data Science** involves **techniques** and **theories** drawn from many fields within the context of *mathematics, statistics, computer science, and information science*.
 Handwritten notes: *computational* (above techniques), *math → science* (above theories), and a blue arrow pointing from the first bullet to the second.
- **Data Science** includes: *data exploration, data cleaning, data analysis, visualization, building predictive models and testing and deploying those models*.
 Handwritten notes: A red box around "building predictive models" with an arrow pointing to "250 terrible predictions" in red. Below this, "1800" is written in green next to a green box divided into three sections. To the left, "23 mt" and "30 mt" are written in red and blue respectively.
- **Data Science** is primarily used to make *decisions and predictions* making use of *descriptive analytics, prescriptive analytics, and machine learning*.
 Handwritten notes: A red box around "machine learning" with a blue scribble inside.
- **Data Science** is about uncovering *findings from data*, and it's about surfacing *insights* that can help *companies* to make *smarter business decisions*.



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- **Data Science process** includes: data exploration, data cleaning, data analysis, visualization, building predictive models, and testing and deploying those models.

garbage

ML

ML Ops

Data Ops



What is Data Science?

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CEO → Chief
CTO → Chief

COO → Chief
CAO → Chief

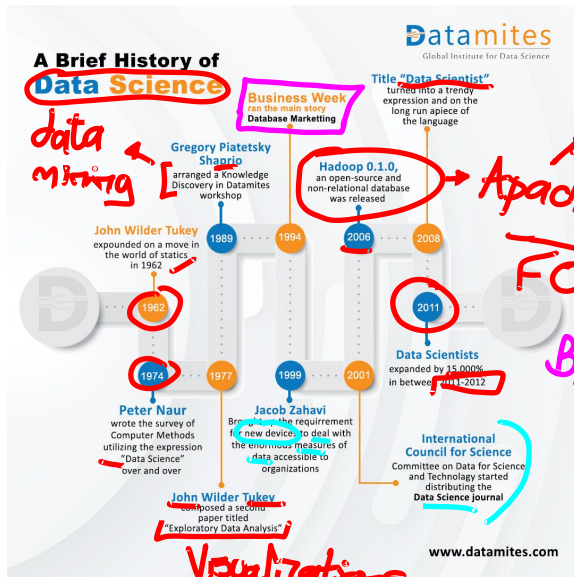


What is Data Science?

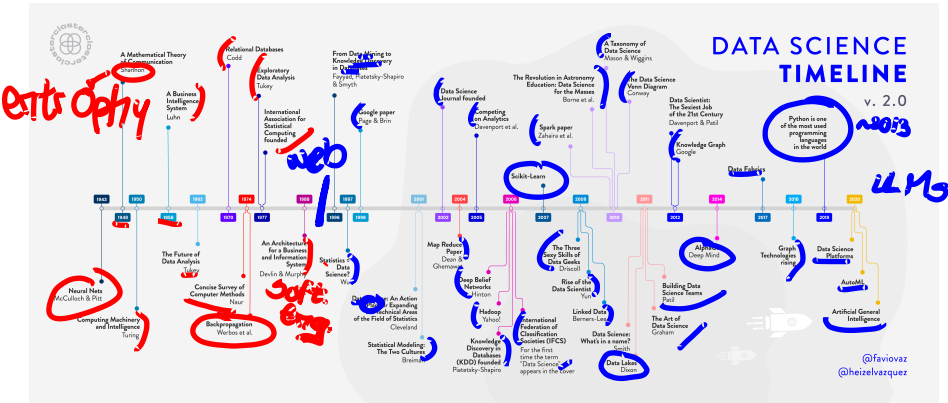
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- **Data Science** is primarily used to **make decisions** and **predictions** making use of *predictive causal analytics, prescriptive analytics, and machine learning*.
- **Data Science** is about **uncovering findings from data**, and it's about surfacing hidden insights that can help enable companies to make smarter **business decisions**.



Brief History of Data Science



Data Science Big Timeline



Data Systems & Big Data

~ 2005

- **Big Data** refers to extremely large data sets that may be analyzed computationally to reveal patterns, trends, and associations, especially relating to human behavior and interactions.
- **Data Systems** are the mechanisms to store, retrieve, and send data. They are crucial for handling big data.

SQL + NoSQL

2004

10 Gb

100

Big Data

500 Gb

Terabytes

2005

2013

Java

→ python → Rumpy



Data Systems & Big Data

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- **Big Data** is characterized by 3Vs: **Volume** (amount of data), **Velocity** (speed of data in and out), and **Variety** (range of data types and sources).

Terabytes

1990
text
↓
format table

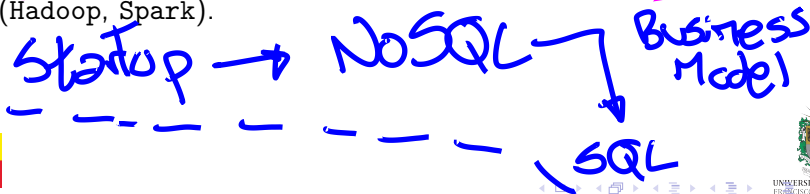
2000
images
video
sound
...

NoSQL
SOAP + XML

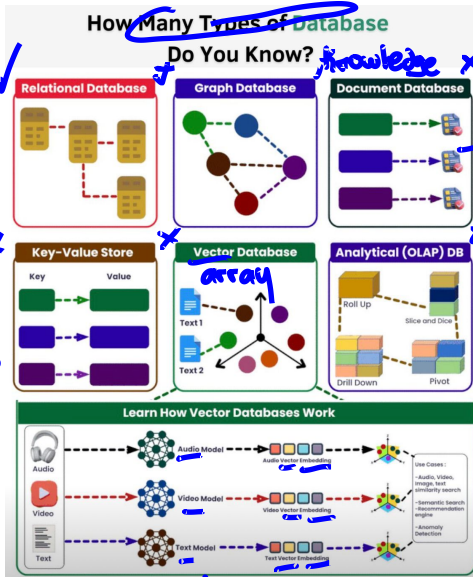


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- **Big Data** is characterized by 3Vs: **Volume** (*amount of data*), **Velocity** (*speed of data in and out*), and **Variety** (*range of data types and sources*).
- **Data Systems** for **Big Data** include **traditional databases** (SQL), **NoSQL databases** (MongoDB, Cassandra), and **distributed systems** (Hadoop, Spark).



Types of Database



JSON
Mongo

Dict
Apache
Redis

Real-Time

RGB
LGB

GPU → Chip AI



Data Lake Vs. Data Warehouse

- **Data Lake** is a storage repository that holds a vast amount of raw data in its native format until it is needed. It is a place to store every type of data in its native format with no fixed limits on account size or file.

- **Data Warehouse** is a system used for reporting and data analysis, and is considered a core component of business intelligence. It is a central repository of integrated data from one or more disparate sources.

- **Data Lakehouse** is a hybrid approach that combines the features of both data lakes and data warehouses. It allows for the storage of raw data like a data lake, while also providing the structured and organized data capabilities of a data warehouse.

CRUD

Google Drive

OneDrive

files

Original format

txt
mpg
mp3
jsonCSV
word
pdf
excel

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/equivalent

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tables → relational databases

ML ← Query → Reports

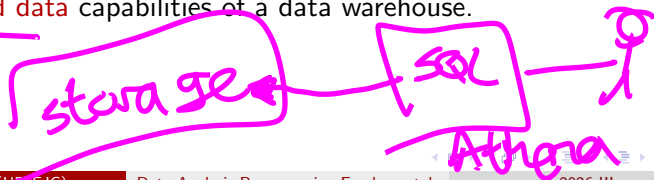


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cloud computing

Fin Ops



Data and MetaData, boom

- **Data** refers to raw, unprocessed, and unorganized facts or details that alone might not make much sense or provide context.
- Metadata is data about data. It provides the *who, what, where, when, why, and how* of the data.
- Examples of metadata include *file size, creation date, modified date, and file type* for a digital file.
- Metadata helps in *data discovery, organization, and interpretation*.
- Metadata is crucial in data management practices like *data governance, data cataloging, and data lineage*.



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Data
Governance



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↓
policies
+
docs

↓
searches
petabytes

data
versions

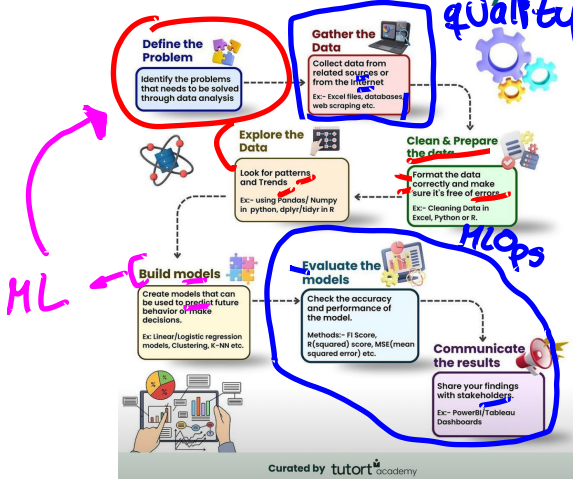
↓
tracking
changes



Process of Data Analysis


[Save for later](#)

Process of Data Analysis



Types of Data Analysis

Different Types of Data Analysis



@alexmartinez



Descriptive Analysis

Summarizes basic data characteristics, like averages, frequencies, and distributions.

Answer "What happened?"



Diagnostic Analysis

Digs deeper, identifying factors influencing specific outcomes or trends.

Answer "Why did this happen?"



Predictive Analysis

Forecasts future events or values based on historical data patterns.

Answer "What might happen next?"



Prescriptive Analysis

Recommends optimal actions based on predicted outcomes and potential impacts.

Answer "What should we do now?"



Other Important Types That Are Widely Used

Exploratory Data Analysis (EDA)

Initial, open-ended investigation to understand data characteristics and uncover potential insights.

Inferential Analysis

Draws conclusions about a larger population based on a sample, using statistical methods to assess confidence levels.

Causal Analysis

Identifies factors causing specific outcomes, often using advanced techniques like causal inference methods.

Mechanistic Analysis

Examines underlying mechanisms and relationships within complex systems, often involving simulation or modeling.

patterns
+
trends

system
analysis

ML



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Artificial Intelligence for Data Science

- **Artificial Intelligence (AI)** is the simulation of human intelligence processes by machines, especially computer systems.
- **AI** involves learning, reasoning, and self-correction.
- **AI** is used in **data science** to *analyze data, make predictions, and automate decision-making processes.*
- **AI** techniques used in **data science** include *machine learning, deep learning, and natural language processing.*

↓
machines who
solve problems
as a human



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Data Science in Industry

- **Data Science** is used in **many industries** to **make decisions** optimize processes, and increase efficiency.
- Data Science is used in **healthcare** to *predict patient outcomes, optimize treatment plans, and personalize medicine.*
- Data Science is used in **finance** to *detect fraud, predict stock prices, and automate trading.*
- Data Science is used in **retail** to *optimize pricing, forecast demand, and personalize marketing.*
- Data Science is used in **manufacturing** to *predict equipment failures, optimize supply chains, and improve quality control.*
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Predictive
Prescriptive



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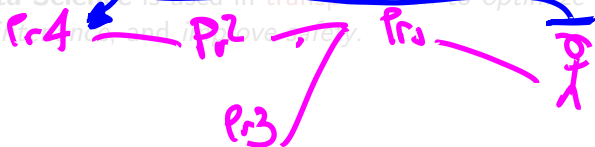
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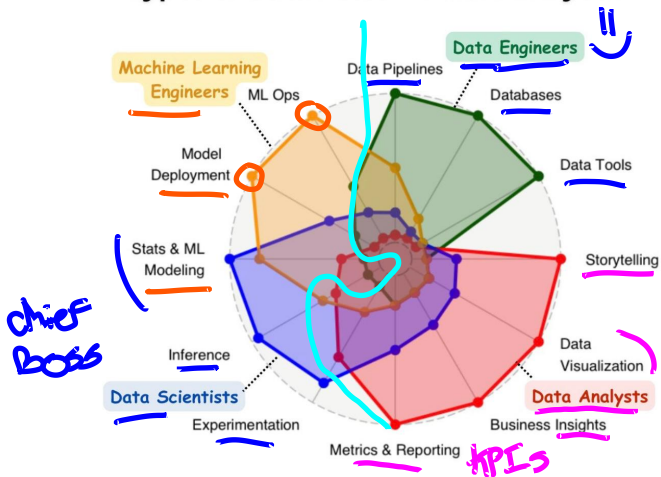
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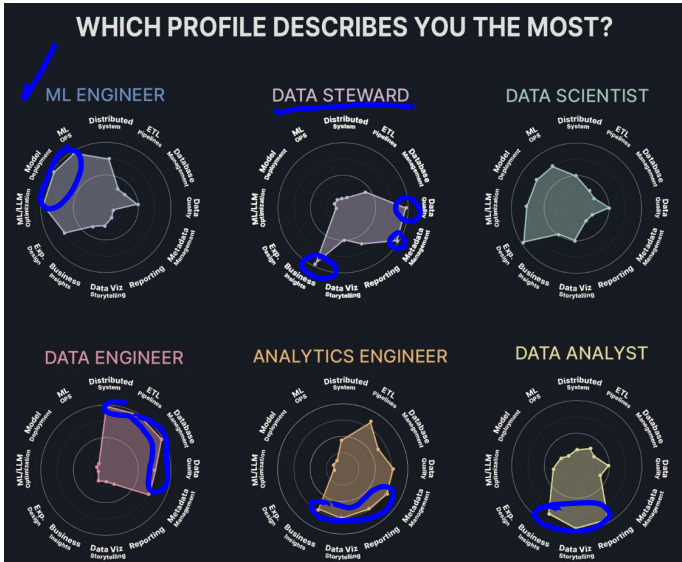


Tech Team — Roles

Types of Data Roles - Where are you?



Tech Team — Data Profiles



Data Scientist Responsibilities

- Collecting large sets of structured and unstructured data from disparate sources. *Terabytes*
- Cleaning and validating the data to ensure accuracy, completeness, and uniformity.
- Analyzing the data to identify patterns and trends.
- Interpreting the data to discover solutions and opportunities.
- Communicating findings to stakeholders using visualization and other means.
- Developing, prototyping, and implementing machine learning models.
- Staying current on techniques and tools in the field, and continually improving skills.



Artificial Intelligence Tech Ecosystem

AI Infrastructure Tools open source

AI FRAMEWORKS, TOOLS & LIBRARIES



AI MODELS



SEARCH



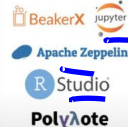
LOGGING & MONITORING



VISUALIZATION



COLLABORATION



@alexwang



Data Science Python Tech Ecosystem

Life is Short, I Use **Python**

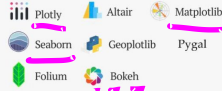


23/11

Data Manipulation



Data Visualization



Statistical Analysis



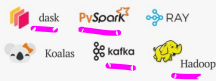
Machine Learning



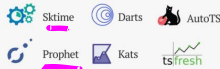
Natural Language Processing



Database Operations



Time Series Analysis



Web Scraping



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming>

